# ADD 334

## MINI PROJECT ZEROTH REVIEW

# Smart Road Early Warning System

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# PROPOSED TOPICS

1. Smart Road Early Warning System
2. AI-Driven Room Measurement & Safety Allocation System
3. Travel Safety and Recommendation System

# INTRODUCTION

- The **Smart Road Early Warning System** is a **mobile-based driver safety application**.
- It helps drivers while **travelling on unfamiliar roads**.
- The system **detects sharp curves** in advance.
- It uses **smartphone sensors** such as GPS, accelerometer, and gyroscope.
- Based on the risk level, the system generates **timely warnings**.
- Alerts are provided through **haptic (vibration) and visual notifications**.
- The goal is to **reduce accidents** by improving driver awareness and reaction time.

# MOTIVATION

- Increase road safety on unknown roads
- Reduce accidents at sharp curves
- Provide real-time alerts without external hardware
- Utilize AI & Data Science for decision making

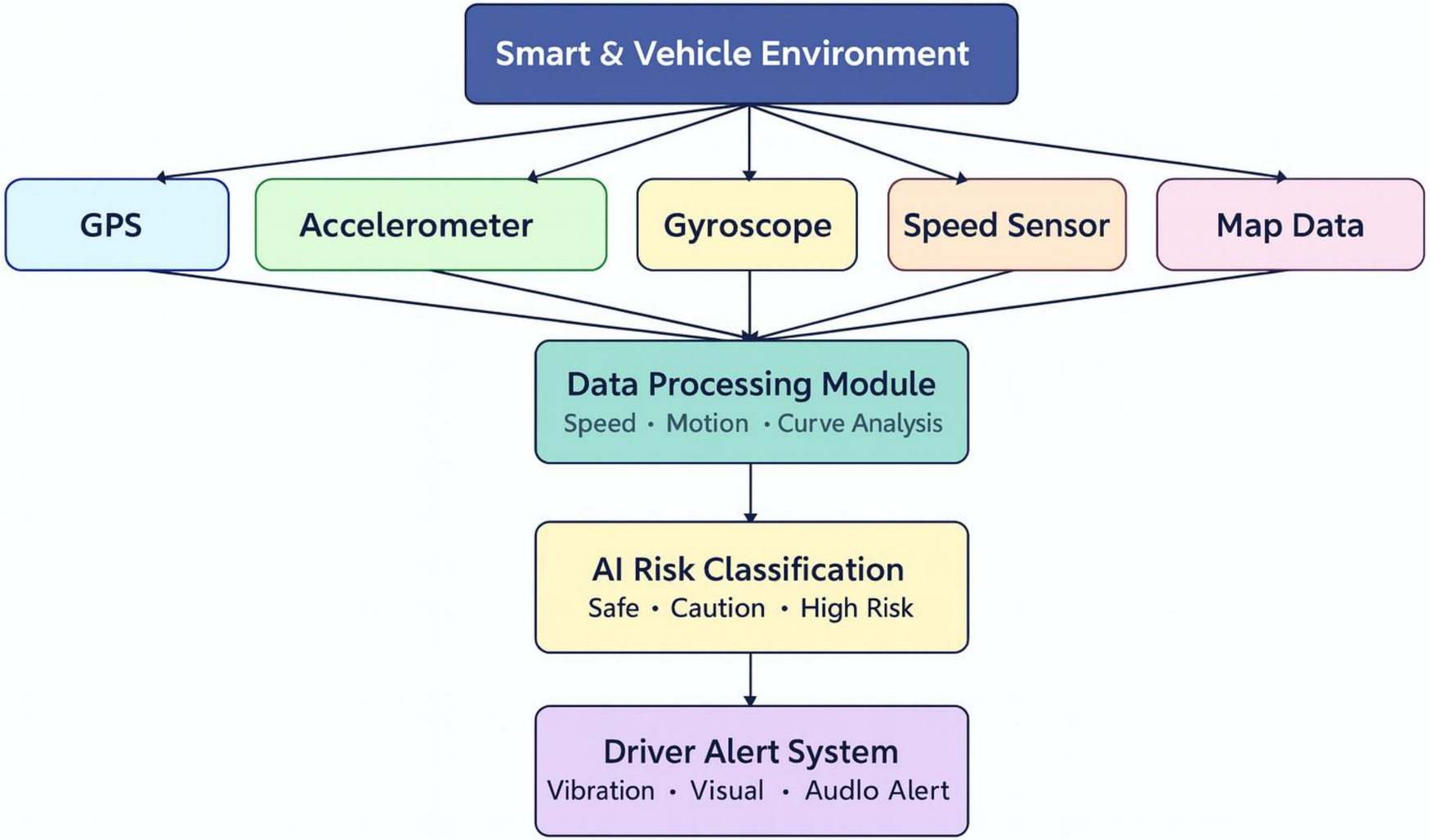
# OBJECTIVE

- Detect dangerous curves in advance
- Warn drivers using vibration alerts
- Classify road segments into safe or risky zones
- Improve driver awareness and reaction time

# SCOPE AND SIGNIFICANCE

- Useful for drivers travelling unfamiliar routes
- Reduces accident risk in hilly and curved roads
- Works using smartphone sensors only
- Scalable for future smart transportation systems

# BLOCK DIAGRAM



# METHODOLOGY

- Collect motion and location data using sensors
- Analyze speed, curve radius and movement patterns
- Apply rule-based and ML classification
- Generate vibration alerts near risky curves

# TECHNICAL FEASIBILITY

- Uses existing smartphone sensors
- Implemented using Python and AI libraries
- No additional hardware required
- Easy integration and scalability

# FINANCIAL FEASIBILITY

- No major hardware cost involved
- Uses smartphone-based implementation
- Economical and cost-effective solution

# SOCIAL FEASIBILITY

- Enhances driver safety
- Reduces accident-related injuries
- Suitable for public and private users
- Improves road safety awareness

# CONCLUSION

- The **Smart Road Early Warning System** offers a **practical and intelligent solution** for improving road safety.
- The system makes use of **AI and Data Science techniques** for risk analysis.
- It utilizes **smartphone sensors** to continuously monitor driving conditions.
- **Timely alerts** are generated when the driver approaches dangerous curves or risky road segments.
- Alerts help drivers **take preventive action in advance**.
- The system supports **safer navigation on unfamiliar roads**.
- Overall, it **reduces the risk of accidents** and enhances driver safety.

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# THANK YOU